

Chacha Chen

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RESEARCH KEYWORDS	Built and deployed AI systems to best support human decisions and augment human intelligence; LLMs for expert tasks; evaluation of LLM reasoning; uncertainty quantification; agents for specialized domain/tasks; synthetic data generation for LLM post-training.
EDUCATION	University of Chicago2025 Ph.D. in Computer Science Advisor: Chenhao Tan Pennsylvania State University2020 Master in Informatics Shanghai Jiao Tong University2019 Bachelor of Science in Computer Science
PROFESSIONAL EXPERIENCE	Apple, AIML ResidentJuly 2025 – Personalized AI for Science (Mentors: Nick Foti and Guillermo Sapiro) - Current LLM evaluations check whether models get the right answer, but not whether they <i>reason correctly to get there</i>, in the intermediate steps. Proposed a new metric for intermediate reasoning quality, grounded in the deviation from optimal Bayesian updates. Scaled evaluation via an efficient vLLM inference pipeline to maximize GPU token throughput, benchmarking ~10 model variants across open-source and closed-source LLMs (1M+ API calls, ~960M tokens, 120 A100 GPUs, ~942 GPU-hours). Found that LLMs systematically over-react to, ignore, or misinterpret evidence, exposing reasoning flaws invisible to existing evaluations. [blog post]. - Built a synthetic data generation pipeline to adapt LLMs to a specialized domain. Designed diverse prompt mixtures via multi-dimensional annotation taxonomy for broad task coverage. Bootstrapped reward/eval function from gold-standard expert responses, combining rubrics evaluation without reference answers with reference-grounded scoring to handle a largely non-verifiable domain, and calibrated against expert ratings. Microsoft Research, Research InternJun. 2023 – Sept. 2023 Evaluation of LLM-powered Generative Search Engines (Mentors: Siddharth Suri, Scott Counts, and Longqi Yang) - Analyzed 80,000 de-identified Bing Copilot conversations to evaluate LLM output quality and emergent use patterns. Developed a task taxonomy and cognitive complexity framework through systematic thematic coding with human validation. - Found that generative search users engage in significantly more knowledge work tasks requiring higher cognitive complexity compared to traditional search, revealing a paradigm shift in search behavior. [publication] Amazon AWS, Applied Scientist InternJun. 2022 – Aug. 2022 Mentors: Sandesh Swamy, Narges Tabari, and Rashmi Gangadharaiah - Developed prefix-tuning method for task-oriented chatbot, enabling adaptive response generation based on conversation context and user intent. IQVIA, Machine Learning Research InternJun. 2020 – Aug. 2020 Mentors: Cao (Danica) Xiao, Fenglong Ma, and Jimeng Sun - Developed UNITE, an uncertainty-aware health risk prediction model with transformer architecture, integrating multi-source clinical data. Improved prediction reliability through uncertainty quantification while preserving predictive performance.
SELECTED PUBLICATIONS	Preprints 1. Chacha Chen*, Matthew Jörke*, Guillermo Sapiro, Sinead Williamson, Nick Foti LLMs as Optimal Information Processors

2. **Chacha Chen***, Yuyang Jiang*, Tong Wu, Liwei Sun, Han Liu, Shi Feng, Chenhao Tan
Collaborative Disagreement Resolution for Scalable Oversight

Conference Papers

- P.1 **Chacha Chen**, Han Liu, Jiamin Yang, Benjamin M. Mervak, Bora Kalaycioglu, Grace Lee, Emre Cakmakli, Matteo Bonatti, Sridhar Pudu, Osman Kahraman, Gul Gizem Pamuk, Aytekin Oto, Aritrick Chatterjee, Chenhao Tan. Human-AI Decision Making with Domain Experts: A Case Study on AI-Assisted Prostate Cancer MRI Diagnosis. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT 2025)*.
- P.2 **Chacha Chen***, Yuyang Jiang*, Dang Nguyen, Benjamin M. Mervak, Chenhao Tan. GPT-4V Cannot Generate Radiology Reports Yet. In *Machine Learning for Health (ML4H 2024)*, *Findings of North American Chapter of the Association for Computational Linguistics (NAACL 2025)*.
- P.3 Yuyang Jiang, **Chacha Chen**, Shengyuan Wang, Feng Li, Zecong Tang, Benjamin M. Mervak, Lydia Chelala, Christopher M Straus, Reve Chahine, Samuel G. Armato III*, Chenhao Tan*. CLEAR: A Clinically Grounded Tabular Framework for Radiology Report Evaluation. In *Findings of the Association for Computational Linguistics (EMNLP 2025)*.
-  P.4 **Chacha Chen***, Shi Feng*, Amit Sharma, Chenhao Tan. Machine Explanations and Human Understanding. *Transactions on Machine Learning Research (TMLR 2023)*; Presented at *ACM Conference on Fairness, Accountability, and Transparency (FAccT 2023)*. **Best Paper Award** at the *ICML Workshop of Human-Machine Collaboration and Teaming*.
- P.4 Vivian Lai, **Chacha Chen**, Alison Smith-Renner, Vera Liao, and Chenhao Tan. Towards a Science of Human-AI Decision Making: A Survey of Empirical Studies. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT 2023)*.
- P.5 Vivian Lai*, Yiming Zhang*, **Chacha Chen**, Vera Liao, and Chenhao Tan. Selective Explanations: Leveraging Human Input to Align Explainable AI. In *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2023)*.
- P.6 Han Liu, Yizhou Tian, **Chacha Chen**, Shi Feng, Yuxin Chen, Chenhao Tan. Learning Human-Compatible Representations for Case-Based Decision Support. In *International Conference on Learning Representations (ICLR 2023)*.
- P.7 **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao. UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data. In *The Web Conference (WWW 2021)*.
- P.8 **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In *Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI 2020)*.
-  P.10 Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD 2020)*.
Best Applied Data Science Paper Award.

AWARDS	Crerar Fellowship, <i>The University of Chicago</i>	Feb. 2021
	AAAI-20 VISA Research Scholarship	Feb. 2020
	Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i>	Jun. 2019
	Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Sep. 2016
COMMUNITY	Organizing committee of FAccT conference 2024, 2025.	
SERVICE		
SKILLS	PyTorch, Transformers, Huggingface, DeepSpeed, vLLM, Python, C++, SQL, R	